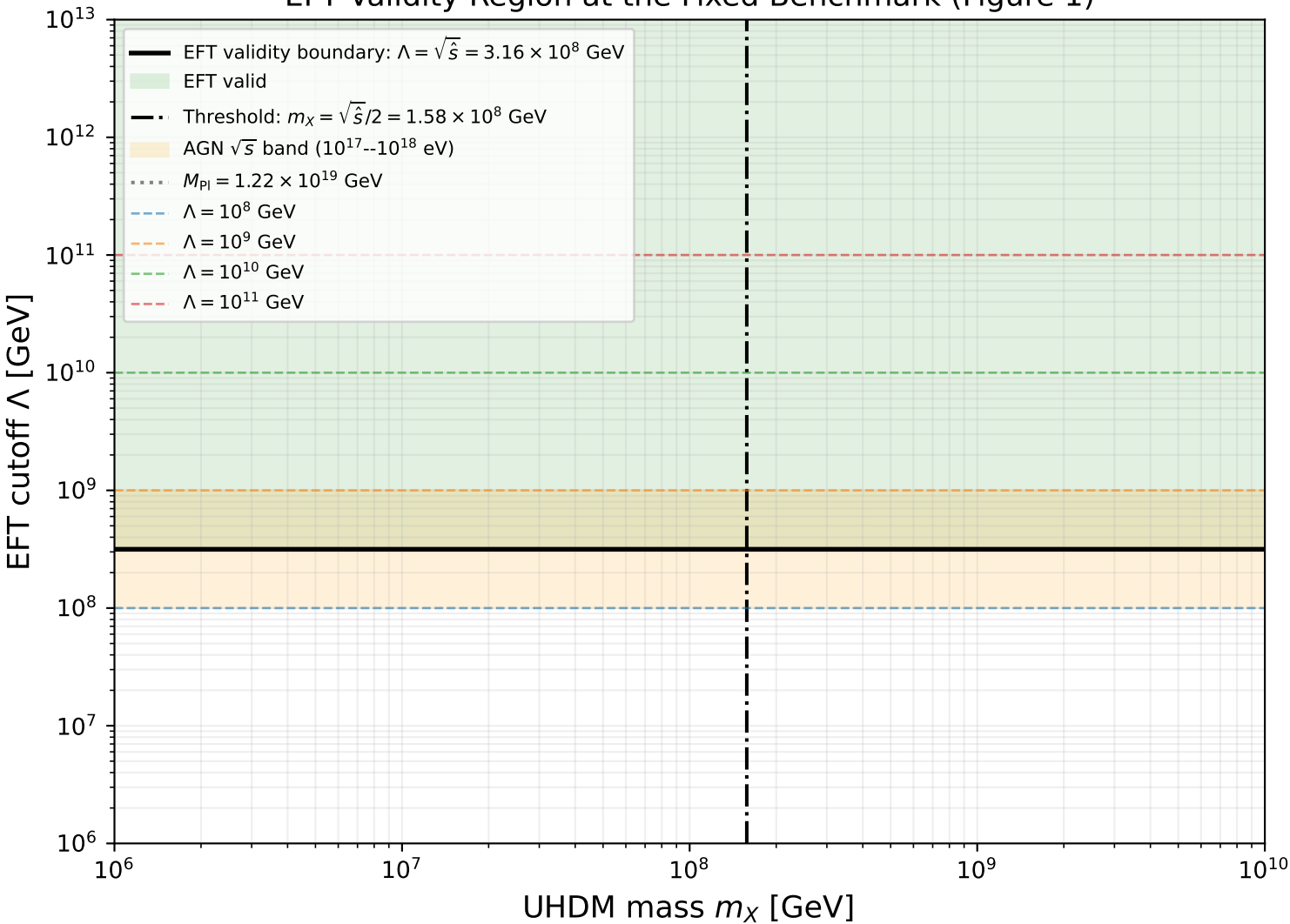
EFT Validity Region at the Fixed Benchmark (Figure 1)
EFT validity boundary: $\Lambda = \sqrt{\hat{s}} = 3.16 \times 10^8$ GeV
EFT valid
Threshold: $m_X = \sqrt{\hat{s}}/2 = 1.58 \times 10^8$ GeV
AGN $\sqrt{s}$ band (10^{17}--10^{18} eV)
$M_{\rm Pl} = 1.22 \times 10^{19}$ GeV
$\Lambda = 10^8$ GeV
$\Lambda = 10^9$ GeV
$\Lambda = 10^{10}$ GeV
$\Lambda = 10^{11}$ GeV
EFT cutoff Λ [GeV]
UHDM mass m_X [GeV]
10^{13}
10^{12}
10^{11}
10^{10}
10^9
10^8
10^7
10^6
10^6
10^7
10^8
10^9
10^{10}